

**Overview of Computer Crimes**

Computer crime also known as cybercrime is when people use computers to do illegal things. It happens when someone who knows a lot about computers uses them in ways that are not allowed. This might include looking at or taking private information that doesn't belong to them. It can also mean damaging other people's computers or files. Sometimes, these people use computers to steal or trick others. The person doing these bad things is often called a hacker. Computer crimes can hurt both the people and the businesses.

**Categories of Computer Crime**

- The computer as a target: The attack seeks to deny the legitimate users or owners of the system access to their data or computers. A Denial-of-Service (a.k.a., DOS or DDOS) attack or a virus that renders the computer inoperable would be examples of this category.
- The computer as an instrument of the crime: The computer is used to gain some information or data which are further used for criminal objective. For example, a hacker may use a computer system to steal personal information.
- The computer as incidental to a crime: Sometimes a computer may not be the primary instrument of the crime; it simply can facilitate it. Money laundering and the trading of child pornography would be examples of this category.
- Crimes associated with the prevalence of computers: This includes the crimes against the computer industry, such as intellectual property theft and software piracy etc.

*Understanding these categories, methodologies and trends is vital for cybersecurity professionals, as it allows them to anticipate threats & develop effective countermeasures.*

**Examples of Computer Crimes**

|                             |                                                                                                                                 |
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| Click fraud                 | Tricking people by clicking on online ads many times to make money or hurt others.                                              |
| Copyright violation         | Using or taking someone else's work (like music, books, or movies) without asking. This is like stealing their ideas.           |
| Cracking                    | Breaking secret codes that keep information safe. This is done to get into private data.                                        |
| Cyber terrorism             | Using computers to scare or threaten people or businesses. This can include hacking or making people pay money to stop attacks. |
| Cyberbully or Cyberstalking | Being mean to people online or following them on the internet in a scary                                                        |

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|                                        | way.                                                                                                                                                        |
| Cybersquatting                         | Taking a web address that should belong to someone else, then trying to sell it to them for a lot of money.                                                 |
| Creating malware                       | Making bad computer programs that can hurt other people's computers or steal their information.                                                             |
| Data diddling                          | Changing numbers or information in computer systems to trick people or steal money.                                                                         |
| Deface                                 | Changing how a website looks without permission. This is like drawing on someone else's house.                                                              |
| DDOS Attack (Denial of Service attack) | Sending so many requests to a computer or website that it can't work properly. It's like too many people trying to enter a shop at once, blocking the door. |
| Data theft                             | Taking other people's private information without their permission. This could be things like passwords or bank details.                                    |
| Doxing                                 | Sharing someone's personal information online without them saying it's okay. This can include things like their address or phone number.                    |
| Espionage                              | Secretly watching a person or company to learn their secrets. It's like being a spy.                                                                        |
| Fake                                   | Things that aren't real but pretend to be. For example, a fake antivirus program that doesn't really protect your computer.                                 |
| Fraud                                  | Tricking people to get money or things. This could be changing bank records or using stolen credit cards.                                                   |
| Green graffiti                         | Using lights or lasers to put pictures or words on buildings without permission. It doesn't leave marks but still changes how things look.                  |
| Harvesting                             | Collecting lots of information about people, like their email addresses or usernames, usually without them knowing.                                         |

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| Human trafficking           | Buying or selling people, which is very wrong and against the law.                                                                                     |
| Identity theft              | Acting like you're someone else to trick people or get things. This could be using their name or personal information.                                 |
| Illegal sales               | Selling/buying things that are not allowed, eg drugs, guns, on the internet.                                                                           |
| Intellectual property theft | Taking ideas or information that someone else made or thought of. This is like stealing someone's thoughts or inventions.                              |
| IPR violation               | Using someone's work, idea, or brand without their permission. This could be copying a song, using a special invention, or making fake brand products. |
| Phishing or vishing         | Tricking people into giving away their private information. This might be done through fake emails or phone calls.                                     |
| Pig butchering              | A tricky way to get people to put money into fake cryptocurrency deals. It often starts with a text message.                                           |
| Ransomware                  | A bad program that locks up someone's computer or files. The criminal asks for money to unlock them.                                                   |
| Salami slicing              | Taking very small amounts of money from many places or people. It's called this because it's like cutting tiny slices from a big salami.               |
| Scam                        | Making people believe something that isn't true, usually to get their money or information.                                                            |
| Sextortion                  | Threatening to share private, often sexual, pictures or information about someone unless they pay money or do what the criminal wants.                 |
| Slander                     | Saying or writing bad things about a person or company that aren't true. This can hurt their reputation.                                               |
| Software piracy             | Using or sharing computer programs without paying for them. It's like making copies of a movie you don't own.                                          |
| Spamming                    | Sending lots of unwanted emails to many people. These are often ads or                                                                                 |

|                            |                                                                                                                                           |
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|                            | scams.                                                                                                                                    |
| <b>Spoofing</b>            | Pretending to be someone else online to trick a computer system. It's like wearing a mask to fool a security camera.                      |
| <b>Swatting</b>            | Making a fake emergency call to send police to someone's house. This is very dangerous and can get people hurt.                           |
| <b>Theft</b>               | Taking things that don't belong to you. This can be computer parts, programs, or information.                                             |
| <b>Typosquatting</b>       | Making a website with a web address that's almost the same as a real one, but with a spelling mistake. This tricks people who type wrong. |
| <b>Unauthorized access</b> | Getting into computer systems you're not allowed to use. It's like breaking into a house you don't own.                                   |
| <b>Vandalism</b>           | Breaking or damaging computer stuff, websites, or other things on purpose.                                                                |
| <b>Wiretapping</b>         | Secretly listening to phone calls or internet talks without permission. It's like eavesdropping, but using technology.                    |

### Overview of Company policy violations

Company policy violations refer to breaches of established guidelines and protocols within an organization, often concerning the use of technology and data protection. A company policy violation in cybersecurity is any employee action that goes against established security rules, leading to potential data breaches, financial loss, and reputational damage. Violations range from malware-prone behaviors like clicking malicious links to negligence like leaving devices unlocked. A company's cybersecurity policy provides the framework to prevent these violations, and consequences for non-compliance can include disciplinary action, reduced system access, or even termination.

#### Common Types of Policy Violations

- **Malicious or Non-Malicious Actions:** Violations can be intentional, such as selling confidential information, or unintentional, like disabling a security configuration.
- **Negligence and Ignorance:** Employees may violate policies due to a lack of awareness or simple negligence, such as opening unknown email attachments, which can introduce malware.
- **Unauthorized Access/Usage:** Connecting personal devices to company networks, accessing social media on work computers, or sharing login credentials are all common policy breaches.

- **Data Handling Issues:** Improperly copying sensitive data to external devices, failing to encrypt files, or not securely disposing of digital media can also be policy violations.

### Consequences of Violations

- **Financial and Reputational Damage:** Data breaches caused by policy violations can lead to substantial financial costs and severely damage a company's credibility with customers and partners.
- **Legal and Regulatory Penalties:** In regulated industries or specific states, companies can face large penalties for inadequate security procedures and data loss.
- **Internal Disciplinary Actions:** Employees caught violating policies can face consequences like additional training, increased system monitoring, suspension of access, or termination.

### How to Address Policy Violations

- **Clear Policies and Training:** Companies must have clear, accessible cybersecurity policies and provide ongoing education to employees about the associated risks.
- **Incident Response Plans:** Establish robust incident response plans to effectively detect, investigate, and mitigate the impact of security incidents caused by policy violations.
- **Regular Audits:** Conduct internal and external audits to ensure compliance and identify potential weaknesses in the security framework.
- **Promote a Culture of Security:** Foster an environment where employees feel comfortable reporting suspicious activity and understand their responsibility in maintaining the company's cybersecurity.

### Preparing a Case

*Preparing a cybersecurity case can mean building a business case for investment or documenting an incident for a legal or internal investigation. For a business case, you need to identify industry threats, assess your company's specific risks, develop a risk-based roadmap for security initiatives, align organizational resources, and demonstrate the return on investment (ROI). For documenting an incident, you need to log the case details, track the attack timeline, identify vulnerabilities, analyze costs, and coordinate with internal teams and external authorities like law enforcement.*

### Preparing a Case for Cybersecurity Investment

This involves a structured argument and is critical for securing resources and confidence in your organization's security.

- **Identify the Need:** Start with a broad overview of industry threats to establish a general need for action.
- **Assess Company Risk:** Use technology-enabled assessments to determine the specific risks and vulnerabilities your organization faces.
- **Develop a Roadmap:** Create a prioritized roadmap of security initiatives, sequencing them based on impact and foundational needs.
- **Align Resources:** Model your organizational structure to support the security strategy and ensure resources are in place.

- Track ROI: Determine and track the return on investment for the proposed cybersecurity measures over time.

### Preparing a Case for a Cyber Incident

This process involves documenting and analyzing a cyber event for various purposes, such as improving incident response or gathering evidence for a legal case.

- Initiate the Case: Log the incident in a system, assigning a unique case number and capturing initial details like title, severity, and impact.
- Identify and Document the Attack: Attacker/Victim: Identify the company that was attacked and the category of attack.
  - Timeline: Define the timeline of events.
  - Vulnerabilities: List specific vulnerabilities exposed during the attack.
- Analyze Costs and Measures: Document the costs associated with the incident and the preventive measures that were taken.
- Correlate and Analyze: Use case management tools to analyze the incident and correlate it with past data to support decision-making.
- Engage with Law Enforcement: Engage with law enforcement early in the process to facilitate quick restoration of operations and increase opportunities for prosecution of cybercriminals.

### Securing Evidence

Securing (digital) evidence in cybersecurity involves a structured forensic process of identification, preservation, collection, and analysis to ensure its integrity, authenticity, and legal admissibility. Key practices include creating forensic copies of data using bit-by-bit imaging, maintaining a meticulous chain of custody, documenting every step, securing evidence in controlled environments, and using specialized tools to capture volatile data without altering the originals. This meticulous approach is crucial for investigating cyberattacks, understanding their scope, and supporting potential legal proceedings.

### The Core Process

- Identification: Recognize the purpose of the investigation and identify the specific digital resources (e.g., computers, networks, mobile devices) that need to be examined.
- Preservation: Isolate the identified devices and data from potential modification or interference.
- Collection: Gather evidence using forensic tools and techniques, focusing on capturing data in a way that maintains its original state.
- Analysis: Examine the collected evidence using specialized tools and processes to identify patterns, trace attack origins, and determine the nature and extent of the incident.
- Documentation & Presentation: Create thorough documentation of all activities and present the findings in a way that is clear and understandable for legal or internal purposes.

- **Forensic Imaging:** Create a bit-by-bit (or bit-wise) copy of the data storage device. This creates an exact replica without altering the original data.
- **Preserve Originals:** Never modify or change data on the original drive or volume. All analysis and examination should be performed on the forensic copies.
- **Chain of Custody:** Meticulously document the handling, transfer, and storage of every piece of evidence. This ensures accountability and prevents tampering.
- **Documentation:** Keep a detailed case log of all actions taken, including dates, times, and methods used.
- **Secure Storage:** Store collected evidence in a secure, access-controlled environment, isolated from networks to prevent tampering or corruption.
- **Minimize Changes:** Minimize any changes made to the data during the collection process by using appropriate tools like write-blockers.
- **Follow Procedures:** Adhere to established cybersecurity policies, procedures, and relevant laws and regulations to ensure evidence remains admissible in legal proceedings.
- **Volatile Data Collection:** Prioritize the collection of volatile data (e.g., RAM, running processes, network connections) before less volatile data, as it can be lost quickly.

### Industrial espionage investigation

A cybersecurity industrial espionage investigation involves detecting, containing, and eradicating threats aimed at stealing proprietary information, trade secrets, or intellectual property using cyber means like malware, phishing, and unauthorized access. Key steps include implementing multi-layered security, employing threat intelligence to identify vulnerabilities, conducting regular security audits, and establishing strong access controls. Investigations often uncover insider threats, necessitating comprehensive employee policies and non-compete agreements to mitigate risks. This can be done by a 3-step process..

#### *1. Detection and Monitoring*

- **Implement robust security measures:** Deploy advanced tools like next-generation firewalls, endpoint detection and response (EDR) systems, and intrusion detection systems (IDS) to monitor network activity and identify suspicious patterns.
- **Utilize threat intelligence platforms:** Use platforms that provide real-time threat intelligence to identify vulnerabilities and active threats before they can be exploited.
- **Monitor user activity:** Track user access to sensitive data and implement zero-trust models to verify identities and limit access to critical resources.
- **Scan for malware and phishing attempts:** Regularly scan for malicious software and analyze email communications for phishing attempts that may be used to gain unauthorized access to systems.

## 2. Incident Response and Containment

- Isolate compromised systems: Disconnect infected or compromised systems from the network to prevent further data exfiltration or lateral movement by attackers.
- Preserve evidence: Collect and preserve digital evidence for forensic analysis without altering it.
- Identify the source of the breach: Investigate to determine the initial entry point, the attackers' methods, and the extent of the damage.

## 3. Investigation and Forensics

- Perform digital forensics: Analyze logs, network traffic, and system data to trace the attacker's actions, the data they accessed, and the methods they used.
- Investigate insider threats: Examine user activity, access logs, and employee behavior to identify any potential involvement of disgruntled employees or other insiders.
- Conduct physical security audits: Assess physical security measures to ensure that digital intrusions are not facilitated by physical access to sensitive areas or equipment.

### Conducting an Investigation

Conducting a cybersecurity investigation involves a methodical, forensic approach to identify, preserve, collect, and analyze digital evidence to understand and respond to a security incident. Key phases include identifying potential evidence, preserving it using forensically sound methods, collecting it, analyzing it with specialized tools and AI, and then creating a report and presenting the findings to authorities. Effective collaboration with law enforcement, other organizations, and international partners is also crucial for building strong cases and addressing borderless cyber threats.

### Gathering evidence

Gathering evidence in a cybersecurity investigation, known as digital forensics, involves the identification, collection, preservation, and analysis of digital traces left by a cyberattack to understand its nature and identify the perpetrators. Key steps include following a strict chain of custody to maintain evidence integrity, which is crucial for admissibility in legal proceedings. This evidence can be found in various digital sources, such as network logs, infected devices, and user activity, and is collected using specialized forensic tools and techniques.

#### The Four Pillars of Digital Forensics

- Identification: Locating potential digital evidence, such as malware files, compromised devices, or malicious scripts, and understanding the scope of the incident.
- Collection: Gathering the identified evidence in a way that doesn't alter or destroy it. This often involves creating forensic copies of hard drives, capturing memory dumps, and collecting network traffic logs.
- Preservation: Isolating the evidence from the live system and protecting it from any further changes. Techniques like hashing are used to create a unique digital signature for each piece of evidence, proving its integrity and authenticity.

- Analysis: Examining the collected evidence to reconstruct the event, identify the root cause, and find traces of the threat actor's activities.

#### Types of Digital Evidence

- Computer Forensics: Evidence from computers, like deleted files, system logs, and user activity.
- Network Forensics: Data from network traffic, including packet captures and firewall logs.
- Mobile Device Forensics: Data from mobile phones, like call logs, text messages, and GPS data.
- Database Forensics: Evidence retrieved from DBs, like transaction records or user access logs.

#### **Bit-stream copy of Evidence**

Bit-stream copy of Evidence is a complete, forensic-grade image of a digital storage medium, ensuring all data, including deleted files and hidden information, is preserved in its original form to maintain the chain of custody and integrity of the evidence. This meticulous process involves creating an exact replica, bit for bit, of the original drive, disk, or memory device, which is then stored on a secure, separate medium for analysis.

#### Why to Use a Bit-Stream Copy?

- Preserves Integrity: It ensures the integrity of the original evidence by creating a duplicate, preventing any potential changes or damage to the volatile source data.
- Maintains Chain of Custody: By working with the bit-stream copy, investigators can keep the original evidence untouched, which is crucial for establishing a legally sound chain of custody.
- Captures Everything: It includes all data, whether visible or hidden, such as deleted files, slack space, and other artifacts that might be relevant to the investigation.
- Allows for Non-Destructive Analysis: The original device can be safely set aside, while analysis is performed on the bit-stream copy, minimizing risks to the live system.

#### The Process of Creating a Bit-Stream Copy

- Forensic Imaging Tools: Specialized software tools like EnCase, FTK, or Autopsy are used to create the bit-stream copy.
- Write-Blocker: A hardware write-blocker is often used to ensure that no data is written back to the original storage device during the copying process, preserving its original state.
- Hashing: Hash values (like MD5 or SHA-1) are calculated for both the original data and the bit-stream copy. If the hash values match, it confirms the copy is an exact, bit-for-bit replica.
- Documentation: The entire process, including the devices used, the software, and the calculated hash values, is thoroughly documented to maintain the integrity of the evidence.

Bit-stream copy is the foundation of digital forensics evidence collection, ensuring that the investigation is built on an accurate and unaltered representation of the digital evidence.

## Storage formats for storing collected digital evidence

The main storage formats for digital evidence are raw images, proprietary forensic formats, and open-source forensic formats, such as the Advanced Forensic Format (AFF).

Raw formats create bit-for-bit copies, while proprietary formats like EnCase (.E01) are vendor-specific but widely used. Open-source formats like AFF are vendor-neutral, offering better interoperability for analysis tools. Choosing a format depends on the needs of the investigation and the available tools.

1. Raw Format: A bit-for-bit copy of a storage device, essentially a direct duplicate of the source data.

- Advantages: Simple, preserves all data, and is compatible with many tools.
- Limitations: Lacks built-in metadata about the evidence or case, making it harder to manage and analyze without additional information.

2. Proprietary Formats: Formats specific to a particular vendor or tool, such as EnCase's E01 format.

- Advantages: Often incorporate features like data compression and integrated metadata (like case information and hash values) to ensure integrity and provide analytical context.
- Limitations: Can lead to vendor lock-in, where only the vendor's tools can fully access or analyze the evidence.

3. Open-Source Forensic Formats: Standardized, non-proprietary formats designed to be used across different forensic tools. The Advanced Forensic Format (AFF) and its variants (like AFF4) are of this category

- Advantages: Promotes interoperability, allowing data to be shared and analyzed by different forensic tools.
- Limitations: May not have the same broad adoption or tool support as some proprietary formats.

Selecting the appropriate format depends on the tools used and the specific requirements of the investigation. Ensuring compatibility with forensic software is vital for effective analysis and reporting.

## Acquisition tools and methods

Cybersecurity data acquisition involves collecting digital evidence using various methods and tools. Organizations must implement proactive data protection policies, continuous monitoring, and employee education to secure their data environment effectively.

Methods for Data Acquisition: determine what data is collected and how it is acquired.

- Logical Acquisition: Focuses on acquiring specific files or folders relevant to an investigation, which is faster and requires less storage than physical acquisition.
- Physical Acquisition: Creates a bit-for-bit copy of the entire storage medium, capturing active files, deleted files, and unallocated space, providing a comprehensive view of the data.

- **Live Acquisition:** Extracts data from a live, operational system to capture volatile information such as running processes, open network connections, and system configurations without shutting down the device.
- **Dead Acquisition (Offline Acquisition):** Collects data from a powered-off or non-operational device, ensuring the integrity of the evidence by preventing changes to the data during the process.
- **Remote Acquisition:** Gathers data over a network connection from remote systems, including cloud-based platforms, eliminating the need for physical access to the device.
- **Memory Forensics:** Involves acquiring and analyzing volatile RAM from live systems to uncover information like running processes, network connections, and encryption keys.

**Tools for Data Acquisition & Security:** used in various aspects of broader cybersecurity operations.

- **Network Analysis: Wireshark:** An open-source tool for capturing and analyzing network protocols in real-time to detect suspicious behavior.
- **Snort:** An open-source network security tool that scans networks for malicious activity and helps prevent intrusions.
- **Disk & Memory Forensics: EnCase:** A comprehensive tool for analyzing hard drives, recovering deleted files, and investigating malware.
- **FTK (Forensic Toolkit):** A fast and efficient tool for scanning and indexing large volumes of data to find evidence.
- **Volatility:** A popular open-source tool for memory forensics, analyzing the contents of RAM.
- **Security Monitoring: Firewalls:** Essential for controlling network traffic and blocking unauthorized access.
- **Intrusion Detection Systems (IDS):** Monitor network and system traffic for unusual or suspicious activity and alert administrators to potential threats.
- **Anti-Malware/Anti-Virus Software:** Detect and remove malicious software from systems.

### **Digital evidence validation :Methods and Tools**

Digital evidence validation in cybersecurity uses multiple methods and various tools /techniques. A robust validation process ensures tools and methods are reliable, traceable, and legally admissible.

#### **Validation Methods**

- **Hash Values:** Calculating and comparing hash values (like MD5 or SHA-256) before and after data collection verifies that the data has not been altered.
- **Cross-Validation:** Using multiple forensic tools and comparing their results helps identify inconsistencies and confirms the accuracy of the analysis.
- **Test Sets/Known Datasets:** Comparing the tool's output against a dataset with known characteristics (test cases) validates the tool's performance.
- **Documentation and Audit Trails:** Thorough documentation of every step, including tool usage and findings, ensures transparency and allows for a complete audit of the forensic process.

- **Specialized Forensic Software:** EnCase and FTK (Forensic Toolkit): Commercial tools used for disk imaging, data recovery, and comprehensive analysis.
- **Autopsy and The Sleuth Kit (TSK):** Open-source platforms and command-line tool suites for various forensic tasks, including disk analysis.
- **Volatility:** A specialized tool for memory forensics, analyzing RAM dumps to uncover running processes and data.
- **Wireshark:** Used for network analysis, capturing and examining network traffic to identify malicious activity.
- **Forensic Imaging:** Creating bit-for-bit copies (forensic images) of digital media to ensure the original data is not altered during analysis.
- **Write Blockers:** Hardware or software devices used to prevent accidental writing of data to the evidence source during the imaging process.
- **Chain of Custody:** A detailed, chronological record of who handled the evidence, when, and why, from collection to presentation, to maintain its integrity.
- **Scientific Validation:** Conducting controlled internal tests and peer reviews based on scientific principles to ensure the tools and methods are reliable and accurate.

### **Storing digital evidence**

*Storing digital evidence securely in cybersecurity involves using protocols, secure media, and access controls to prevent tampering and unauthorized access. This is quite crucial for maintaining evidence integrity and ensuring its admissibility in legal proceedings.*

### **Key Principles for Storing Digital Evidence**

- **Preserve Integrity:** The primary goal is to ensure the evidence is unaltered and can be used in an investigation or court.
- **Maintain Chain of Custody:** Document every step of the evidence's lifecycle, including who handled it, when, and how it was stored.
- **Isolate from Networks:** Store evidence on systems not connected to live networks to prevent remote access or corruption.
- **Utilize Secure Storage:** Use physical data safes or secure cloud solutions designed to protect against physical theft and environmental hazards.
- **Implement Access Controls:** Limit who can access the evidence and maintain logs of all access attempts and actions taken.
- **Encrypt the Data:** Encrypt storage devices and cloud services to add a layer of security against unauthorized access.
- **Use Tested Tools and Methods:** Employ only tools and methods that have been validated for accuracy and reliability in digital forensics.

## Evidence Retention

Evidence retention involves the proper collection, secure storage, and careful tracking of digital data from a security incident to ensure its integrity for future analysis; working with the principle of "collect first, act second" to avoid compromising evidence.

### *Why Evidence Retention is Crucial*

- **Incident Analysis & Response:** Preserved evidence provides the foundation for understanding the full scope of an attack, identifying attacker methods, and effectively containing and eradicating threats.
- **Legal & Regulatory Compliance:** Properly collected and preserved evidence is essential for legal investigations and can help an organization meet data regulations and standards.
- **Forensic Investigations:** Evidence is vital for in-depth analysis, allowing investigators to reconstruct events and identify the root cause of a cyber incident.
- **Building Trust & Reputation:** Demonstrating a commitment to proper evidence handling can build trust with customers and partners and protect the organization's reputation.

### Key Steps in Evidence Retention

- **Identify Evidence Sources**
  - Determine all potentially relevant sources of data, including servers, workstations, network devices, mobile devices, and cloud services.
- **Collect Evidence**
  - **Forensic Imaging:** Create bit-for-bit duplicates of drives to preserve the data without altering the original system.
  - **Live Data Collection:** Capture volatile data from RAM, network traffic, and system logs, which are critical for understanding current activity.
  - **Documenting the Scene:** Photograph the physical location of devices and document surrounding items to provide context.
- **Preserve and Store**
  - **Chain of Custody:** Maintain a detailed, unbroken record of who collected, handled, and stored each piece of evidence to ensure its integrity and admissibility.
  - **Secure Storage:** Store evidence in isolated, secure locations with strict access controls to prevent unauthorized access or tampering.
  - **Hashing:** Use cryptographic hash functions to verify the authenticity of evidence and detect any alteration.
- **Analysis**
  - **Work on Duplicates:** All analysis should be performed on the forensic images or duplicates, never on the original devices, to avoid any modification of the evidence.

### **CASE STUDY: Familiarizing Autopsy for Windows - a free forensics tool.**

Autopsy is a free, open-source, graphical forensic platform for Windows that analyzes digital media, such as disk images, by allowing users to extract and review data like deleted files, browser history, and metadata.

#### **Key Features**

- **Free and Open-Source:** Autopsy is available at no cost, allowing for widespread use in academic and professional settings.
- **User-Friendly Interface:** It provides a graphical interface to simplify complex digital forensic tasks.
- **Cross-Platform Compatibility:** While known for its Windows performance, it can analyze disk images from different operating systems.
- **Modular Design:** Its plug-in architecture enables customization with community-developed or custom modules.
- **Comprehensive Analysis:** Autopsy can identify file types, extract artifacts like email and browser data, perform keyword searches, and view file metadata.

The Autopsy website offers official documentation, tutorials, and a forum to help new users get started. We shall do this Case Study / Practice in our NCERC MCA Lab.



**AUTOPSY**  
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<https://www.autopsy.com/download/>

#### **Getting Started with Autopsy**

- **Download and Install:** Obtain the Autopsy installer from its official website.
- **Create a Case:** Launch Autopsy and create a new case, providing details like the case name and examiner's name.
- **Add a Data Source:** Add a disk image or other data source to your case.
- **Run Analysis:** Autopsy will automatically process the data, running various analytical modules.
- **Review Results:** Explore the processed information through the interface, including file views, a timeline, and keyword hits.
- **Generate Reports:** Create detailed reports in formats such as HTML, PDF, or Excel to share findings.

Prepared by

**Dr. Sudheer S Marar**  
Professor & HOD – MCA  
Dean Academics - NCERC